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Unit – 4 : Session – 4 : SLO – 2 | Tree Traversals — Post-Order Traversal

■ I. Algorithm for Post-Order Traversal (Left, Right, Root)

- Step 1: Repeat Steps 2 to 4 while TREE != NULL
- Step 2: POSTORDER(TREE -> LEFT)
- Step 3: POSTORDER(TREE -> RIGHT)
- Step 4: Write TREE -> DATA
- [END OF LOOP]
- Step 5: END

■ II. Post-Order Traversal — Tree 1 (root 41)

- Tree structure: 41 -> left 30 (left 23; right 37 -> right 38); right 79 (left 45; right 81 -> right 96)
- Post-order: 23, 38, 37, 30, 45, 96, 81, 79, 41

■ III. Post-Order Traversal — Tree 2 (root 73)

- Tree structure: 73 -> left 39 (left 26 -> right 31; right 43 -> right 67); right 93 (left 73, right 95)
- Post-order: 31, 26, 67, 43, 39, 73, 95, 93, 73

■ IV. True or False

1. Preorder + Postorder enough to reconstruct the original tree? FALSE — preorder and postorder do not uniquely
 - determine a binary tree (unless it is a full binary tree); an inorder sequence is needed.
2. Postorder + Inorder enough to reconstruct the original tree? TRUE — the last element of postorder is the root;
 - inorder splits left/right subtrees; recurse to rebuild the unique tree.
3. Leaf nodes appear in the same relative order in preorder, inorder and postorder? TRUE — leaves are always
 - visited left-to-right in all three depth-first traversals.